

VeriTrust AI

Enterprise AI Governance & Risk Intelligence Platform

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Target Domain Exposure: BFSI, Healthcare, HR & Employment, Aviation & Aerospace

1. Live Cloud Production Deployment

VeriTrust AI is fully deployed in production as a decoupled cloud platform on Vercel and Render:

Component	Platform	Production URL	Status
Frontend UI	Vercel CDN	https://veri-trust-ai-governance.vercel.app	Active ■
Backend API	Render Web Service	https://veritrust-ai-governance-1.onrender.com	Active ■
Swagger Docs	Render OpenAPI	https://veritrust-ai-governance-1.onrender.com/docs	Active ■
Health Check	Render REST Endpoint	https://veritrust-ai-governance-1.onrender.com/api/health	Active ■
GitHub Repo	GitHub Public	https://github.com/vanshika-data-lab/VeriTrust-AI-Governance	Active ■

2. Executive Summary & Questionnaire Alignment

- **Avoids 'Black-Box Prompting':** Replaces arbitrary LLM outputs with a deterministic, mathematically verifiable 10-dimension risk scoring matrix (scoring_matrix.py) cross-referenced against codified regulatory articles.
- **Mandatory 6-Tier Evidence Classification:** Classifies citations across (1) Law/Regulation, (2) Regulatory Guidance, (3) Industry Standard, (4) Academic Research, (5) Vendor Information, and (6) General Web Content.
- **Live 'Surprise Record' Evaluator Test:** Accepts completely novel, un-seeded AI use cases dynamically at runtime with instant 4-step synthesis.
- **Persistent SQLite Database Store:** Features structured relational persistence (governance_app.db) with cascading foreign keys for all assessments and citations.
- **100% Free & Open-Source Stack:** Runs with zero paid API keys or commercial license dependencies (FastAPI, React 18, Vite, SQLite3).

3. 5-Layer Enterprise AI Architecture

Layer	Technologies	Key Functional Responsibilities
Layer 1: UI Layer	React 18, Vite, Lucide, Vercel CDN	Executive Risk KPIs, 10-Area Progress Visualizers, Dynamic Test Form, 6-Tier Explorer, Report Exporter.
Layer 2: App/API Layer	FastAPI 0.110, Uvicorn, Render	High-speed stateless REST endpoints, CORS middleware, Pydantic v2 validation, health checks.
Layer 3: AI Intelligence	Python 3.10, Scoring Matrix, RAG	10-dimension deterministic risk scoring, metadata enrichment, statutory obligation synthesis.
Layer 4: Data Layer	SQLite3 Relational Database	ACID-compliant storage for use cases, assessments, dimensional scores, citations, and knowledge base.
Layer 5: Research Layer	HTTPX Async Client, 6-Tier KB	15 canonical statutory reference documents + live public web crawler for surprise use cases.

4. Deterministic 10-Dimension Risk Scoring Matrix

Every AI system is evaluated across 10 mandatory governance dimensions with explicit statutory linkages:

Governance Dimension	Weight	Governing Law / Canonical Standard	Mitigating Controls Enforced
1. Data Lineage & Quality	10%	EU AI Act Art. 10 / ISO 42001 A.8	Provenance tracking, distribution audit
2. Privacy & Data Protection	10%	GDPR Art. 35 DPIA / FTC Guidance	PII pseudonymization, DPIA review
3. Bias & Demographic Fairness	15%	NYC LL144 / Title VII 80% Rule	Disparate impact ratio, proxy filtering
4. Human Oversight Controls	15%	EU AI Act Art. 14 / NIST GOVERN 1.2	Emergency kill-switch, override logging
5. Explainability & Recourse	10%	EOA 12 CFR 1002.9 / GDPR Art. 22	SHAP/LIME attribution, adverse notices
6. Cybersecurity Robustness	10%	SOC 2 CC6.1 / NIST AI RMF Manage	Prompt injection firewalls, red-teaming
7. Decision Impact Severity	10%	ISO/IEC 23894 / Microsoft RAIA	Executive sign-off, automated rollback
8. Regulatory Exposure	10%	EU AI Act High-Risk Annex III	Regulatory conformity technical dossier
9. Model Reliability & Drift	5%	Stanford HAI / NIST MEASURE	RAG grounding confidence, OOD filters
10. Continuous Monitoring	5%	ISO/IEC 42001 Clause 6.1	Real-time telemetry, SIEM alerting

5. 6-Tier Regulatory Evidence Classification Corpus

Source Tier	Canonical Authorities & Frameworks Included in Knowledge Base
Tier 1: Law / Regulation	EU AI Act (Reg 2024/1689), GDPR (Reg 2016/679), NYC Local Law 144, Equal Credit Opportunity Act (ECOA Reg B)
Tier 2: Regulatory Guidance	NIST AI Risk Management Framework 1.0, FTC Fair Information Guidance, EEOC Disparate Impact Guidance
Tier 3: Industry Standard	ISO/IEC 42001:2023 (AIMS), IEEE 7000-2021 Ethics in Systems Design, Microsoft Responsible AI Standard v2
Tier 4: Academic Research	Stanford Center for Research on Foundation Models (CRFM), MIT Disparate Impact Algorithmic Benchmarks
Tier 5: Vendor Information	Model System Cards & Evaluation SLAs, Cloud Provider AI Compliance Audits, API Security Specs
Tier 6: General Web Content	HTTPX Live Regulatory Search Engine, Audited Industry Trackers, Legal Commentary Portals

6. REST API Specification (FastAPI)

Method	Endpoint	Description
GET	/api/health	Health check & service liveness ping.
GET	/api/use-cases	List all pre-seeded and dynamically tested AI systems.
DELETE	/api/use-cases/{id}	Delete a specific use case with cascading database removal.
GET	/api/assessments/{id}	Retrieve full 10-dimension assessment with 6-tier citations.
POST	/api/assess	Dynamically evaluate any novel surprise AI system.
GET	/api/sources	Search 6-tier knowledge base with query and tier filters.
GET	/api/analytics	Retrieve executive metrics, risk breakdown, & tier distribution.
GET	/api/export-report/{id}	Export structured JSON compliance audit report.

7. Evaluator Demonstration Workflow (5 Steps)

1. **Overview:** Open <https://veri-trust-ai-governance.vercel.app>. Observe executive KPIs, 10-area visual meters, and 6-tier classification.
2. **Knowledge Base:** Navigate to 6-Tier Knowledge Base. Filter by Law/Regulation to inspect EU AI Act, GDPR, and NYC LL144 canonical summaries.
3. **Dynamic Surprise Record Test:** Go to Dynamic 'Surprise Record' Test. Load a preset or input novel use case and click Execute Assessment.
4. **Detailed Audit & Evidence Review:** Inspect the synthesized score, EU AI Act High-Risk rating, dimensional breakdown, and open the 6-tier Citation Drawer.
5. **Compliance Report Export:** Click Export Full Audit Report to download the JSON compliance audit archive.